IT3011 - Theory & Practices in Statistical Modelling

Two Population Means

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Properties of Expected Values

Consider X and Y are random variables follows a probability distribution.

$$E(X+Y) = E(X) + E(Y)$$

$$E(X-Y) = E(X) - E(Y)$$

$$E(kX) = kE(X)$$

Where k is a constant value.

$$E(aX \pm bY) = aE(X) \pm bE(Y)$$

Properties of Variances

Consider X and Y are random variables follows a probability distribution and they are **independent**.

$$V(X+Y) = V(X) + V(Y)$$

$$V(kX) = k^2 V(X)$$

$$V(X-Y) = V(X+(-Y)) = V(X) + V(-Y) = V(X) + (-1)^2 V(Y) = V(X) + V(Y)$$

Where k is a constant value.

$$V(aX \pm bY) = a^2 V(X) + b^2 V(Y)$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Case A - $\sigma_1^2 \neq \sigma_2^2$ but known

Let Y_1 and Y_2 be independent random variables distributed as $N(a, b)$ and $N(c, d)$ respectively. Then, $Y_1 \pm Y_2 \sim N(a \pm c, b + d)$.

Using the above property,

Let $x_1, x_2, x_3, \ldots, x_n$ is a random sample from from $N(\mu_1, \sigma_1^2)$ and $y_1, y_2, y_3, \ldots, y_m$ is a random sample from from $N(\mu_2, \sigma_2^2)$. We know that,

$$\bar{X} \sim N\left(\mu_1, {\sigma_1^2}/{n}\right) \qquad \bar{Y} \sim N\left(\mu_2, {\sigma_2^2}/{m}\right)$$

So,

$$(\bar{X} - \bar{Y}) \sim N\left(\mu_1 - \mu_2, {\sigma_1^2}/{n} + {\sigma_2^2}/{m}\right)$$

So,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{{\sigma_1^2}/{n} + {\sigma_2^2}/{m}}} \sim N(0,1)$$

(1−α)100% Confidence Interval for the difference between two normally distributed population means.

Then we can construct (1−α)100% Confidence Interval for $\mu_1 - \mu_2$ as follows,

$$P\left(-Z_{\frac{\alpha}{2}} \leq Z \leq Z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$P\left(-Z_{\frac{\alpha}{2}} \leq \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n + \sigma_2^2/m}} \leq Z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

So the (1−α)100% Confidence Interval for $\mu_1 - \mu_2$ is,

$$(\bar{X} - \bar{Y}) \pm Z_{\frac{\alpha}{2}}\sqrt{\sigma_1^2/n + \sigma_2^2/m}$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Case B - $\sigma_1^2 = \sigma_2^2$ but known

Let $x_1, x_2, x_3, \ldots, x_n$ is a random sample from from $N(\mu_1, \sigma^2)$and $y_1, y_2, y_3, \ldots, y_m$ is a random sample from from $N(\mu_2, \sigma^2)$. We saw that for $\sigma_1^2 \neq \sigma_2^2$,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n + \sigma_2^2/m}} \sim N(0,1)$$

When $\sigma_1^2 = \sigma_2^2 = \sigma^2$,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n + 1/m}} \sim N(0,1)$$

So the confidence interval can be derived as,

$$(\bar{X} - \bar{Y}) \pm Z_{\frac{\alpha}{2}}\left(\sigma\sqrt{1/n + 1/m}\right)$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Case C - $\sigma_1^2 = \sigma_2^2$ but unknown

Let $x_1, x_2, x_3, \ldots, x_n$ is a random sample from from $N(\mu_1, \sigma^2)$and $y_1, y_2, y_3, \ldots, y_m$ is a random sample from from $N(\mu_2, \sigma^2)$. We saw that for $\sigma_1^2 \neq \sigma_2^2$,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n + \sigma_2^2/m}} \sim N(0,1)$$

When $\sigma_1^2 = \sigma_2^2 = \sigma^2$,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n + 1/m}} \sim N(0,1)$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

We also know that,

$$\frac{(n-1)S_1^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

$$\frac{(m-1)S_2^2}{\sigma^2} \sim \chi^2_{(m-1)}$$

Let Y_1 and Y_2 be independent random variables distributed as $\chi^2_{(a)}$ and $\chi^2_{(b)}$ respectively. Then, $Y_1 + Y_2 \sim \chi^2_{(a+b)}$.

Using the above property,

$$W = \frac{(n-1)S_1^2}{\sigma^2} + \frac{(m-1)S_2^2}{\sigma^2} \sim \chi^2_{(m+n-2)}$$

Because $\frac{(n-1)S_1^2}{\sigma^2}$ and $\frac{(m-1)S_2^2}{\sigma^2}$ are independent.

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Using the definition of t distribution,

$$T = \frac{Z}{\sqrt{\frac{W}{n+m-2}}} \sim t_{(n+m-2)}$$

$$T = \frac{\frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{\sigma\sqrt{1/n+1/m}}}{\sqrt{\frac{\frac{(n-1)S_1^2}{\sigma^2}+\frac{(m-1)S_2^2}{\sigma^2}}{n+m-2}}} = \frac{\frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{\sqrt{1/n+1/m}}}{\sqrt{\frac{(n-1)S_1^2+(m-1)S_2^2}{n+m-2}}} = \frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{S_p\sqrt{1/n+1/m}} \sim t_{(n+m-2)}$$

Then we can construct $(1-\alpha)100\%$ Confidence Interval for $\mu_1-\mu_2$ as follows,

$$P\left(-t_{(n+m-2),\frac{\alpha}{2}} \le T \le t_{(n+m-2),\frac{\alpha}{2}}\right) = 1-\alpha$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

So the $(1-\alpha)100\%$ Confidence Interval for $\mu_1 - \mu_2$ is,

$$(\bar{X} - \bar{Y}) \pm t_{(n+m-2),\frac{\alpha}{2}} \; S_p\sqrt{{}^{1}/_{n} + {}^{1}/_{m}}$$

If n and m are very large (say 30), the $(1-\alpha)100\%$ Confidence Interval for $\mu_1 - \mu_2$ can be written as,

$$(\bar{X} - \bar{Y}) \pm Z_{\frac{\alpha}{2}} \, S_p\sqrt{{}^{1}/_{n} + {}^{1}/_{m}}$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Case D - $\sigma_1^2 \neq \sigma_2^2$ but unknown

The $(1-\alpha)100\%$ Confidence Interval for $\mu_1 - \mu_2$ is,

$$(\bar{X} - \bar{Y}) \pm t_{(df),\frac{\alpha}{2}} \sqrt{S_1^2/n + S_2^2/m}$$

$$df = \frac{\left(S_1^2/n + S_2^2/m\right)^2}{\frac{\left(S_1^2/n\right)^2}{n-1} + \frac{\left(S_2^2/m\right)^2}{m-1}}$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

Ex-

From a large class 4 students were drawn and their Maths marks were 64, 66, 89 and 77. From a second large class 3 students were drawn and their Maths marks were 56, 71 and 53. Assuming the classes are normally distributed and have equal variances, find 95% confidence interval for the difference between the mean of the two Maths marks.

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

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$$\bar{X} = 74 \; and \bar{Y} = 60$$

$$S_1^2 = 132.67 \text{ and } S_2^2 = 92$$

$$S_p = \sqrt{\frac{3 \times 132.67 + 2 \times 92}{4 + 3 - 2}} = 10.78$$

$$t_{2.5\%,5} = 2.57$$

$(1-\alpha)100\%$ Confidence Interval for the difference between two normally distributed population means.

$$(\bar{X}-\bar{Y}) \pm t_{(n+m-2),\frac{\alpha}{2}}\ S_p\sqrt{{}^{1}/_{n}+{}^{1}/_{m}}$$

$$14 \pm 10.78\sqrt{\frac{1}{4}+\frac{1}{3}} \times 2.57$$

$$[-7.034, 35.09]$$

Comparing two population means.

Consider following hypotheses,

$$H_0 : \mu_1 - \mu_2 = k \text{ , } H_1 : \mu_1 - \mu_2 \neq k$$

$$H_0 : \mu_1 - \mu_2 = k \text{ , } H_1 : \mu_1 - \mu_2 > k$$

$$H_0 : \mu_1 - \mu_2 = k \text{ , } H_1 : \mu_1 - \mu_2 < k$$

Consider $\mu_{def} = \mu_1 - \mu_2$,

$$H_0 : \mu_{def} = k \text{ , } H_1 : \mu_{def} \neq k$$

$$H_0 : \mu_{def} = k \text{ , } H_1 : \mu_{def} > k$$

$$H_0 : \mu_{def} = k \text{ , } H_1 : \mu_{def} < k$$

This is pretty much similar to one population case. Only difference can be seen in the test statistic.

When we compare two population means we have to check where the null hypothesis is $\mu_1 = \mu_2$ situation which is $\mu_1 - \mu_2 = \mu_{def} = 0$ (k=0)

Comparing two population means – Unpaired case.

Let $x_1, x_2, x_3, \ldots, x_n$ is a random sample from from $N(\mu_1, \sigma_1^2)$and $y_1, y_2, y_3, \ldots, y_m$ is a random sample from from $N(\mu_2, \sigma_2^2)$. We know that,

$$\bar{X} \sim N\left(\mu_1, {\sigma_1^2}/{n}\right)$$

$$\bar{Y} \sim N\left(\mu_2, {\sigma_2^2}/{m}\right)$$

So,

$$(\bar{X} - \bar{Y}) \sim N\left(\mu_1 - \mu_2, {\sigma_1^2}/{n} + {\sigma_2^2}/{m}\right)$$

This is the sampling distribution of difference between two sample means. Then we can use **z** or **t** for known or unknown population variances respectively as the test statistic. Following sub cases explain the test statistic for different conditions of population variances.

Comparing two population means – Unpaired case.

Case A - $\boldsymbol{\sigma_1^2 \neq \sigma_2^2}$ but known

Then the test statistic is,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n + \sigma_2^2/m}} \sim N(0,1)$$

Case B - $\boldsymbol{\sigma_1^2 = \sigma_2^2}$ but known

Then the test statistic is,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n + 1/m}} \sim N(0,1)$$

Comparing two population means – Unpaired case.

Case C - $\boldsymbol{\sigma_1^2 = \sigma_2^2}$ but unknown

Then the test statistic is,

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p\sqrt{{}^{1}/_{n} + {}^{1}/_{m}}} \sim t_{(n+m-2)}$$

Where,

$$S_p = \sqrt{\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}$$

Comparing two population means – Unpaired case.

Case D - $\boldsymbol{\sigma_1^2 \neq \sigma_2^2}$ but unknown

Then the test statistic is,

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n + S_2^2/m}} \sim t_{(df)}$$

Where,

$$df = \frac{\left(S_1^2/n + S_2^2/m\right)^2}{\frac{\left(S_1^2/n\right)^2}{n-1} + \frac{\left(S_2^2/m\right)^2}{m-1}}$$

Comparing two population means – Unpaired case.

Ex-

A test was given to a group of 100 scouts and 144 guides. The mean score for scouts was 27.53 and the mean score for the guides was 26.81. Assuming a common population standard deviation of 3.48, test at 5% level of significance whether the scouts performance in the test was better than that of the guides. Assume that the scores are normally distributed.

Comparing two population means – Unpaired case.

Ex-

A test was given to a group of 100 scouts and 144 guides. The mean score for scouts was 27.53 and the mean score for the guides was 26.81. Assuming a common population standard deviation of 3.48, test at 5% level of significance whether the scouts performance in the test was better than that of the guides. Assume that the scores are normally distributed.

$$n_1 = 100 \qquad n_2 = 144$$

$$\bar{X} = 27.53 \qquad \bar{Y} = 26.81$$

$$\sigma = 3.48$$

$$H_0 : \mu_1 - \mu_2 = 0 \text{ , } H_1 : \mu_1 - \mu_2 > 0$$

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n + 1/m}} \sim N(0,1)$$

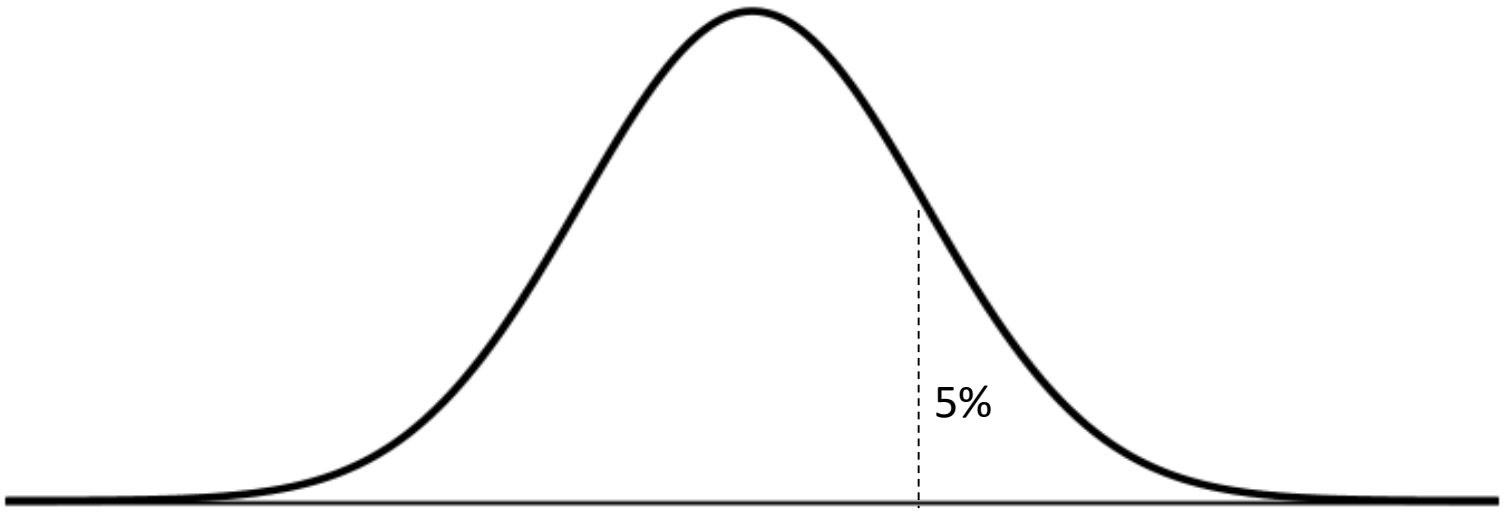
Comparing two population means – Unpaired case.

Under H_0,

$$Z_{cal} = \frac{(27.53 - 26.81) - (0)}{3.41\sqrt{{}^{1}/_{100} + {}^{1}/_{144}}} = 1.5894$$

This is one tail right tail test.

$Z_{cal} < Z_{critical}$



$Z_{critical} = 1.645$

Comparing two population means – Unpaired case.

Do not reject H_0 5% level of significance.

Conclusion - **There is no evidence to say that the scouts performance in the test was better than that of the guides at 5% level of significance.**

Comparing two population means – Paired case.

The objective of this section is testing the difference between population means based on means of two dependent samples.

Paired samples: The sample selected from the first population is related to the corresponding sample from the second population.

Assumptions we made under paired samples:

The difference of the pairs follow a normal distribution or the number of pairs is large. (note here that if the number of pairs is less than 30, we need to check whether the **differences** are normally distributed, but we do not need to check for the normality of each population.)

The key to analyze a matched sample design is to use the difference data.

Comparing two population means – Paired case.

Consider hypothesis,

$$H_0 : \mu_{def} = k \text{ , } H_1 : \mu_{def} \neq k$$

$$H_0 : \mu_{def} = k \text{ , } H_1 : \mu_{def} > k$$

$$H_0 : \mu_{def} = k \text{ , } H_1 : \mu_{def} < k$$

Test statistic:

Let d=differences between the pairs of data, then $\bar{d}$=mean of these differences. Then the test statistic is,

$$T = \frac{\bar{d} - \mu_{def}}{S_{def} \big/ \sqrt{n}} \sim t_{(n-1)}$$

Where n denotes the number of pairs or the number of differences.

Comparing two population means – Paired case.

Ex-

Trace metals in drinking water affect the flavor and unusually high concentration can pose a health hazard. Ten pairs of data were taken measuring zinc concentration in bottom water and surface water. Does the data suggest that the true average concentration in the bottom water exceeds that of surface water? (5% level of significance)

	Location									
	1	2	3	4	5	6	7	8	9	10
Zinc concentration in bottom water	0.430	0.266	0.567	0.531	0.707	0.716	0.651	0.589	0.469	0.723
Zinc concentration in surface water	0.415	0.238	0.390	0.410	0.605	0.609	0.632	0.523	0.411	0.612

Comparing two population means – Paired case.

Since the sample size is not large enough (Less than 30), we need to check whether the differences follow a normal distribution.

	B	S	D
0	0.430	0.415	0.015
1	0.266	0.238	0.028
2	0.567	0.390	0.177
3	0.531	0.410	0.121
4	0.707	0.605	0.102
5	0.716	0.609	0.107
6	0.651	0.632	0.019
7	0.589	0.523	0.066
8	0.469	0.411	0.058
9	0.723	0.612	0.111

Thus, we can test and show that the difference may come from a normal distribution

Comparing two population means – Paired case.

$$H_0 : \mu_{def} = 0 \text{ , } H_1 : \mu_{def} > 0$$

$$T_{cal} = 4.86$$

$$t_{0.05,9} = 1.833$$

Since $T_{cal} > t_{0.05,9}$, Reject H_0 at 5% level of significance.

Conclusion - **There is an evidence to say that the true average concentration in the bottom water exceeds that of surface water at 5% level of significance.**